Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/30/2016
PI Name:	Hammond, Timothy M.D.		
Project Title:	Evolution of Genotypic and Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique to Microgravity		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 7/15/2017, per NSSC information (Ed., 7/18/16) NOTE: End date is now 7/15/2016, per NSSC information (Ed., 4/20/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Birdsall, Holly (Baylor College of Medicine) Nislow, Corey (Stanford University/University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)		
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Task Description:

This application investigates microbial growth and physiological responses to the multiple stimuli encountered in space flight environments, using Saccharomyces cerevisiae as a model system. Yeast are an ideal model because their complete gene sequence is known; there are extensive and sophisticated clone libraries available for study; and they can easily be flown in space, held at +4oC in a quiescent state, and activated by simple warming. Regulatory mechanisms in yeast are highly similar to mammalian cells. We have shown that microgravity activates a unique set of genes in yeast, but the specific molecular mechanisms remain incompletely defined.
In a corollary set of experiments, we are utilizing next generation sequencing techniques to assess the effects of long term exposure to radiation on the yeast model organism Saccharomyces cerevisiae. The cellular response to DNA damage has been studied extensively in a variety of model systems. The most comprehensive view of this process has come from yeast which, owing to the conservation of this essential pathway, shares homologs of their human DNA repair counterparts. A key observation from our group and others is that different types of DNA damage require unique repair genes and proteins. Accordingly, we are studying the cellular response to energetic particle radiation experienced in space using whole genome sequencing of thousands of yeast cells exposed to a large matrix of space-specific conditions.

We hypothesize that by identifying specific regulatory motifs within the promoter regions of microgravity-responsive genes, we will be able to determine transcription factors that bind the motifs, and thus the signaling pathways involved in microbial growth and physiological responses. We propose ground-based studies with transition to a future flight experiment, designed so as to have a high probability of obtaining genome wide data.

The proposed experiments are timely because (a) recent improvements in promoter analysis have vastly increased the accuracy of promoter elements predictions; (b) novel yeast clones allow direct measurement of gene expression by fluorescence using the plate reader in the Advanced Biological Research System (ABRS) that is already in place on the International Space Station (ISS); (c) yeast on agar plates can be assayed repetitively, thereby eliminating sample to sample variability to increasing the statistical power of the data; (d) new reagents allow design of experiments with minimal up mass (less than 2 kg); and (e) all data can be returned by telemetry, eliminating the need for down mass.

To test our hypothesis, we will:

1) Identify promoter motifs associated with genes responding to Rotating Wall Vessel (RWV)-emulated microgravity compared to static cultures. The RWV will be used to emulate microgravity as we have already demonstrated that it induces changes in yeast gene expression that are similar to those of yeast flown in space. The Yeast Green Fluorescent Protein (GFP) clone collection will permit identification, by changes in fluorescence, of genes whose expression changes in response to real and simulated microgravity. After extensive normalization and analytic procedures, microgravity-responsive genes will be grouped into clusters on the basis of similar kinetics, under the assumption that these are likely to be co-regulated. We will then search the upstream sequence of co-expressed genes and identify candidate binding motifs for transcription factors.

2) Confirm the role of regulatory sequences in directly activating transcription of a set of genes changed by RWV-simulated microgravity and to identify transcription factors that interact with these regulatory motifs. This specific aim makes novel use of the yeast deletion collection, a systematic gene-by-gene set of complete gene deletions, the only such resource that exists for any group of organisms. Transcription factors will be identified by robotically introducing a reporter plasmid that codes for 1) a suicide-inducing enzyme that is activated in all clones except the one(s) with a deletion in the gene for the transcription activating factor that binds to the promoter motif or 2) an endogenously activated essential gene to identify repressors.

3) Confirm and extend the results obtained in our ground based studies on the selective growth pressure in the true microgravity environment. Utilizing the genome wide yeast GFP clone collections spotted on agar plates, we propose a parallel flight study. Our question can ultimately only be definitively answered in the true microgravity of space, because on the ground gravity must always be onset with an equal and opposite force.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This works aims to understand the genetic control mechanism that controls the gene expression response to cellular responses to microgravity in yeast.
Yeast have 70% homolog with the human genome, and are a useful model system to define primary and secondary drug metabolism pathways that cannot be done in current mammalian cell culture systems.

The short term aim is to map the genetic promoter elements that mediate gene responses to microgravity. The long term goal is to use this information to unveil drug metabolism pathways in the unique metabolic environment provided by microgravity.

We propose to establish a sustained genomics platform on the ISS to understand the long term effects of the radiation, and other features of the ISS environment, on crew health by dissecting the essential processes of DNA damage and DNA repair. This has broad Earth benefits in understanding effects and optimization of medically necessary radiation therapy and chemotherapy with alkylating agents.

The APEX-02-2 work quantitatively measures radiation damage to yeast DNA exposed to space radiation.. Using state of the art technologies, the science team can, for the first time, conduct a highly powered, genome wide analysis of mechanisms of radiation damage in space. Knowledge gained from this investigation will build upon previous data to understanding the mechanisms of radiation damage, and may provide simple approaches to ameliorating space based radiation damage.

By using yeast strains to understand the effects of space based radiation stresses at the cellular level, a greater knowledge of the regulatory mechanisms at work in cells will be gained. A hallmark of radiation stresses is that they induce physiological responses; primary among them are DNA damage. As yeast is a model eukaryotic organism, DNA damage changes caused by radiation could lead to a greater understanding of these processes on Earth, thus benefiting all

	citizens. Knowledge gained from this investigation will build upon previous data to understanding the mechanisms of radiation damage, and may provide simple approaches to ameliorating clinical radiation damage.
Task Progress:	In the first phase of this project wildtype and radiation sensitive stains of Saccharomyces cerevisiae were flown in space and where resident on the ISS for 30 and 180 days before recovery (SpaceX4 and SpaceX6). We used 100 independent yeast strains to survey the effects of radiation on this model organism, and collected whole genome sequence data for 1000 genomes on space-based and ground controls. This cohort comprises 100 individual strains, 90 of which contain verified deletion alleles in a variety of conserved pathways including MEC1, MRE11, and RAD4, 5, 7, 18, 24, 27, 27, and RAD51 mutants, encoding essential proteins including checkpoint kinases, helicases, nucleases, and ligases. These 1000 genomes provide a baseline spectrum of mutations against which we will calibrate additional experimental samples. As a testament to the robustness of this platform, a replicate set of samples originally intended to return on SpaceX4 after 30 days of exposure actually remained on the ISS for 180 days, returning on SpaceX6. This allowed us to double the size of our data set and provided insight into the effects of long-term exposure. The yeast libraries were constructed and sequencing is complete. Early analysis of the results has been reported in abstract form. We are dissecting the essential processes of DNA damage and DNA repair. Whole Genome Sequencing (WGS) of thousands of strains collected over hundreds of generations has uncovered space-specific vulnerabilities, and pathway analysis is allowing analysis of known and novel radiation sensitive pathways. The data sets will be publicly accessible on open-source, community databases including Genelab, as soon as current analysis is complete. Beginning in 2005, developments in massively parallel sequencing Next Generation Sequencing or NGS changed the way in which DNA sequencing was performed. This revolutionized the field of genomics by delivering a complete human genome sequences in a fraction of the time (days versus years) and at a fraction of the cost (thousands of dollars versus billions of dollars). No aspect of life sciences has been untouched by this revolution and new fields have emerged, such as ecological genomics, microbiome genomics, and personalized genomics. The microbiome (all the microbes present in diverse body sites) is becoming a key diagnostic of the health of the host. Epi-genomics, the study of DNA modifications, including those that promote serious diseases, is now possible, as are the delineation of complex traits that can determine susceptibility to disease and offer insights on treatment. The recent Decadal Survey Report, “Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era,” by the National Research Council (http://www.nap.edu/), specifically mentions the need for NASA to utilize new Omics research technologies. The biologic effects of radiation in space continue to pose a challenge to space exploration and Omics may provide useful new tools to apply to this problem. The aim of the second stage of this project is to fly a yeast payload to understand the molecular mechanisms of radiation damage, and whether radiation repair mechanisms have unique aspects in space. Multiple genome-wide series of deletion clones will be assayed to study the genomics of radiation effects. We propose to return the yeast arrays from space and analyze the DNA by next generation sequencing and make comparison to ground controls. Pathway analysis will then allow analysis of known and novel radiation sensitive pathways. The data sets will be placed on publicly accessible websites, including GeneLab, as a community resource. Phase I of the experiments validated the use of high-throughput, next-generation sequencing (NGS) to assess the effects of long term exposure to radiation using the yeast, Saccharomyces cerevisiae. In our experience, 100 unique deletion alleles will provide a robust dataset for reconstructing pathways that are both directly and indirectly involved in the cells resistance to radiation damage. Multiple genome-wide series of ~4,800 deletion clones of Saccharomyces cerevisiae of “Baker’s yeast” has been prepared for use in the reflight. To translate this study data to clinical and space health countermeasures, 10 new deletion series have been created with a second putative control genes also deleted. That is, ten yeast strains have been constructed by genetic deletion, each lacking a putative radiation repair control gene. The entire yeast deletion series of ~4,800 deletion clones is re-derived using the strains with putative control gene deletion as the background strain in which all other deletions are constructed, resulting in genome wide pools of double deletion strains. This fresh approach aims to determine the molecular mechanisms of radiation damage to facilitate understanding radiation damage, and may provide simple approaches to ameliorating space based and clinical radiation damage.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Other Journals or Periodicals	Hammond TG, Stodieck L, Birdsall HH, Becker JL, Koenig P, Hammond JS, Gunter MA, Allen PL. "Effects of microgravity and clinorotation on the virulence of Klebsiella, Streptococcus, Proteus, and Pseudomonas." Gravitational and Space Research. In press as of June 2016. , Jun-2016
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Articles in Peer-reviewed Journals	Hammond TG, Allen PL, Birdsall HH. "Validation of assays for reactive oxygen species and glutathione in Saccharomyces cerevisiae during microgravity simulation." Gravitational and Space Research. 2015 Dec;3(2):42-53. http://gravitationalandspacebiology.org/index.php/journal/article/view/701/732 , Dec-2015
Articles in Peer-reviewed Journals	Hammond T, Allen P, Birdsall H. "Is there a space-based technology solution to problems with preclinical drug toxicity testing?" Pharm Res. 2016 Jul;33(7):1545-51. Review. http://dx.doi.org/10.1007/s11095-016-1942-0 ; PubMed PMID: 27183841 , Jul-2016
Awards	Hammond T. "Distinguished International Visiting Fellow Paeter Wall Institute of Advanced Studies, University of British Columbia, September 2015." Sep-2015